# Disk Failure: From Prediction to Decision under Cost Trade-offs

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***Abstract*—Proactive disk replacement is a core AIOps task, yet disk failure predictors are usually judged with row-level accuracy metrics that ignore what operators need: early warning, a bounded number of alerts, and a positive net cost. This proposal studies disk failure prediction on the public Backblaze Drive Stats dataset (2025, about 115 million drive-days and 4.3 thousand failures) from an operational viewpoint. We (1) quantify how much actionable warning SMART[1] signals really provide, (2) compare classification, survival and remaining-useful-life formulations under a fixed alert-budget evaluation, and (3) derive calibrated, cost-aware alert thresholds and show how they shift with the ratio between replacement and failure costs. The expected outcome is a reproducible evaluation protocol and decision guidelines for storage operators.**

## I. Problem Statement

Hard disk drives (HDD) and Solid State Drives (SSD) remain the backbone of large-scale storage, and their failures cause degraded arrays, costly rebuilds and, in the worst case, risks to data durability. Predicting failures from the S.M.A.R.T. telemetry that drives report about themselves, so that at-risk drives are replaced or drained before they fail, is a classic AIOps problem [1], [2], [3].

However, models are usually judged by row-level accuracy. It means that each row is one drive on one day, and the model is scored on predicting its label. It uses accuracy, precision/recall or AUC. Row-level scoring ignores the other levels at which a predictor is actually used. For instance, drive-level answers the question "was a failing drive flagged at least once before it died?", and event-level answers "was each failure flagged early enough to act on?".

In that context, row-level metrics do not capture what an operations team needs: (i) *lead time*, the number of days between the first alert for a drive and its failure, since a warning is useless if it arrives too late to act; (ii) a *bounded alert volume*, since technicians can inspect only a few drives per day and too many alerts cause alert fatigue; and (iii) *cost*, since a false alarm wastes a drive and labor while a missed failure triggers an emergency intervention. With annualized failure rates of 1–2% [4], [5], accuracy is misleading, and a default 0.5 threshold ignores costs entirely [6].

The right trade-off is also not universal: it depends on the relative cost of an unnecessary replacement and of an unplanned failure, which vary across operators, redundancy schemes and over time. The question is therefore no longer "can we predict failures?" but "which alerts should we act on, and when?"

This project is linked to LOG6309E through its themes of incident prediction, AIOps, data analytics for software engineering, and DevOps for ML applications: time-based validation, probability calibration, cost-aware decision making and reproducible pipelines.

## II. Research Questions

**RQ1:** How much actionable warning do SMART signals provide, and for which failures?

*Motivation.* No predictor can beat the signal in the data, and field studies found that a notable share of failures show no SMART warning [5]. Knowing this ceiling per manufacturer and model shows operators where prediction can and cannot help.

*Approach.* We align failed drives on "days before failure" and detect, for each one, the first change in critical attributes that we chose (SMART 5, 187, 188, 197, 198)[2] with thresholds and change-point detection, compared with matched healthy drives. We report the share of failures with at least $L \in \{3, 7, 14\}$ days of warning versus silent failures, by manufacturer, model and capacity.

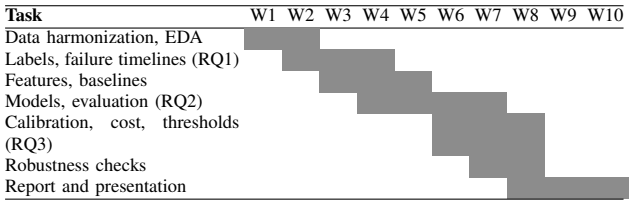
**RQ2:** Which problem formulation and feature set gives the best operational outcome under a fixed alert budget?

*Motivation.* On the one hand, formulations are usually compared by AUC, which hides differences at the top of the ranking, where decisions are made. On the other hand, Feature choice matters in practice [3], yet it is judged by average prediction quality rather than by its effect on lead time and alert volume.

*Approach.* On identical time-based and drive-level splits, we compare (a) classification of "fails within $H \in \{7, 14, 30\}$ days", (b) survival (time-to-event) models and (c) remaining-useful-life regression, each turned into a daily risk score. Baselines are the rule "any critical counter $> 0$", which reflects common threshold-based practice [2], [5], and logistic regression, a standard interpretable classifier [1]. We also ablate snapshot versus rolling/delta features. We replay 2025

[1] Self-Monitoring, Analysis, and Reporting Technology. They track internal drive metrics to detect early signs of hardware failure.

[2] 5: reallocated sectors; 187: reported uncorrectable errors; 188: command timeouts; 197: current pending (unstable) sectors; 198: offline uncorrectable sectors. We chose them because they are raw event counters that stay at zero on healthy drives, they are comparable across vendors, and prior field studies identified them as the strongest failure indicators [4], [7]. SMART 187 and 188 are not reported by all manufacturers, which is handled by our per-manufacturer analysis.




day by day, alerting on the top $K$ drives, and report actionable recall (failures alerted at least $L$ days ahead), median lead time and unique drives alerted per week, with bootstrap confidence intervals over drives.

**RQ3:** Which alert threshold best balances unnecessary replacements and unexpected failures, and how does it shift when the cost ratio changes?

*Motivation.* Choosing a threshold is a cost decision: a low threshold causes needless replacements, a high one leaves the fleet exposed.

## III. Dataset

We use the public Backblaze Drive Stats dataset [4]: a daily snapshot of every operational drive, stored as one CSV per day with one row per drive (date, serial number, model, capacity, a binary failure flag, and raw and normalized SMART attributes). Quarterly archives are available; Q1 2025 has 90 daily files (10.9 GB uncompressed), and the full year 2025 covers about 115.6 million drive-days with 4,317 failures (AFR 1.36%) [4]. Data through 2026 also exists and can serve as an optional future-period test.

*Feasibility.* The data is open and already local. We built a CSV-to-Parquet pipeline with DuckDB and started the EDA on Q1 2025 before scaling to the full year. Known challenges and our handling: (i) rare positives and small per-model counts, handled with exposure filters and bootstrapping; (ii) the failure flag appears only on a drive's last day, and some drives vanish without failing (retirement), which we treat as censored or exclude; (iii) schema changes between quarters, handled by column harmonization; (iv) vendor-specific SMART semantics, handled with per-manufacturer analysis; and (v) a single provider's cold-storage workload, partly mitigated by leave-one-model-out and forward-chaining validation.

## IV. Related Work

Mahdisoltani et al. [1] predict drive errors in the field to enable proactive actions, and Xu et al. [2] forecast disk errors in a large cloud by combining SMART with system-level signals and ranking disks by risk. Lu et al. [3] improve prediction on a large production fleet through feature and model selection. An earlier large-scale field study [5] shows that SMART is informative but insufficient on its own.

These works mainly optimize prediction quality. Our project differs in four ways: (1) drive-level operational evaluation with a minimum lead time and an alert budget; (2) classification, survival and RUL formulations compared on equal footing; (3) an explicit cost model whose optimal threshold is derived, calibrated and stress-tested under varying cost ratios, building on cost-sensitive learning [6]; and (4) a fully open, reproducible protocol on public 2025 data.

## V. Time Planning of Project

The project spans ten weeks (W1–W10), to be aligned with the official course schedule. The proposal presentation precedes W1, a mid-project check is planned around W5, and the final presentation and the 8–10 page report fall in W9–W10.

## References

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[3] S. Lu, B. Luo, T. Patel, Y. Yao, D. Tiwari, and W. Shi, "Making disk failure predictions SMARTer!" in *Proc. 18th USENIX Conf. File and Storage Technologies (FAST)*, 2020.

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